Dart Programming Language

Data Type : NUMBER

num
- int → Whole Number → 12, 17, 20, 36.
- double → Decimal Number → 10.5, 13.8, 19.1.

1) round() → 12.7 → 13 (Int)
   → -8.5 → -9

2) roundToDouble() → 12.7 → 13.0
   — -8.5 → -9.0
   (double)

3) ceil() → 12.2 → 13 (Int)
   -4.1 → -4

4) ceilToDouble() → 12.2 → 13.0
   → -4.1 → -4.0
   (double)

5) floor() → 12.9 → 12 (Int)
   -4.7 → -5

6) floorToDouble() → 12.9 → 12.0
   -4.7 → -5.0
   (double)

7) truncate() → 12.2 → 12 (Int)
   -4.9 → -4

8) truncateToDouble() → 12.2 → 12.0
   -4.9 → -4.0
   (double)

9) abs() → -12.1 → 12.1

10) Clamp(_ , _)
   120
   Clamp(0, 100) → 100
   -10 → 0

Arithmetic operators

(+ , - , * , / , ~/ , %)

```
print(10 % 3);   // 1
Print(10.remainder(3));   // 1
```

11) toStringAsFixed(2)

9.2634568...

12) toStringAsPrecision(4)

9.263

13) print(10.bitLength);   1010

// 4;

14) print(12.gcd(18));

// 6

15) int population = 1_000_000;

print(population); // 1000000;

bitwise op

5 & 3 , 5 | 3, 5 ^ 3, ~5 , 5<<1,

5>>1

import 'dart:math';

16) print(min(10,20)); // 10

17) Print(max(10, 20); // 20

18) double value = 1.5e+3;

print(value); // 1500.0

$= 1.5 \times 10^{+3}$

$= 1.5 \times 1000$

$= 1500.0$

print(value.toStringAsExponential(1));

1.5e+3

19) int Color = 0xFF;

Print(color); // 255

10 - A

11 - B

12 - C

13 - D

14 - E

15 - F

F F

$16^2 + 16^1 \times 15 + 16^0 \times 15$

240 + 15

= 255

print(color.toRadixString(16));

// ff

20)

y = $\log_e 10$;

e^y = 10;

print(log(10)); // 2.3028....

double y = log(10);

print(exp(y).round()); // 10

21) sin, cos, tan

pi = 180°

print(sin(pi/2)); // 1

print(cos(0)); // 1

print(tan(pi/4)); // 1

Type casting

String → Int/Double

Int → Double

Double → Int

I/D → String

tryParse() ✓

Int.Parse() ✓

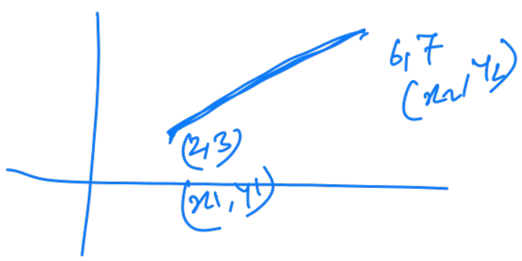
22)

r = 5 unit

area = πr^2

```
void main() {
double radius = 5;
double area = pi * radius * radius;
double area = pi * pow(radius, 2);
print(area);
}
```

23)



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

```
void main() {
    double x1, y1, x2, y2;
        x1 = 2;
        y1 = 3;
        x2 = 6;
        y2 = 7;
    double distance = sqrt(pow(x2-x1, 2)
                         + pow(y2- y1, 2));
        print(distance);
```

24) 5-digit OTP

```
print(Random().nextInt(90000) + 10000);
```

Random().nextInt(100); + 10

0 - 99

0+1 - 99+1

10 - 109

0 — 89999

10000 → 99999

25) Roller Dice

```
print(Random().nextInt(6) + 1);
```

0 - 5

1 - 6

26) Coin Toss :

```
print(Random().nextBool() ? "Head" : "Tail");
```

Homework

1. Age = ✓

Human Age = Dog Age × 7

1 = 7 ; ✓

2. inch → height = ____

foot, inch ✓

3. radius = 7

area (A) = πr^2 ✓

(S) = $2\pi r$ ✓